

INTRODUCTION

There is a wide diversity in living organisms in our biosphere. Now a question that arises in our minds is: Are all living organisms made of the same chemicals, i.e., elements and compounds? You have learnt in chemistry how elemental analysis is performed. If we perform such an analysis on a plant tissue, animal tissue or a microbial paste, we obtain a list of elements like carbon, hydrogen, oxygen and several others and their respective content per unit mass of a living tissue. If the same analysis is performed on a piece of earth's crust as an example of non-living matter, we obtain a similar list. What are the differences between the two lists? In absolute terms, no such differences could be made out. All the elements present in a sample of earth's crust are also present in a sample of living tissue. However, a closer examination reveals that the relative abundance of carbon and hydrogen with respect to other elements is higher in any living organism than in earth's crust.

Analysis of Chemical Composition

Analysis of organic compounds - Chemical analysis is done to find out the types of organic compounds (compounds containing carbons) found in living tissues.

- Living tissue taken.
- Grinded in trichloro acetic acid (TCA) (Cl_3CCOOH) to obtain slurry.
- Slurry is filtered to get filtrate (acid-soluble fraction: contains biomicromolecules) and retentate (acid insoluble fraction or pellet contains biomacromolecules)
- Thousands of organic compounds found in filtrate.
- Separation techniques used for separating one compound from another.
- Molecular formula and probable structure found by using analytical techniques.
- All carbon-containing compounds derived from living organism are called **biomolecules**.

NOTE - the sum total composition of acid soluble and acid insoluble fraction represents the entire composition of cellular pool.

Analysis of inorganic compounds:

- Living tissue taken.
- It is dried to evaporate all water, and the remaining material gives its dry weight.
- The dried material is burnt.
- All organic and gaseous compounds will be removed to leave "ash".
- Ash contains many inorganic compounds like Ca, Mg, S, P, etc.

Component	Formula
Sodium	Na^+
Potassium	K^+
Calcium	Ca^{++}
Magnesium	Mg^{++}
Water	H_2O
Compound	$NaCl, CaCO_3, PO_4^{3-}, SO_4^{2-}$

Table: A List of Representative Inorganic Constituents of Living Tissues

Table: A Comparison of Element Present in Non-living and Living Matter

Element	% weight of	
	Earth's crust	Human body
Hydrogen (H)	0.14	0.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.60	65.0
Nitrogen (N)	Very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	Negligible

Biomolecules

- All the carbon compounds obtained from living tissues are called **biomolecules**.
- Chemistry point of view: Functional groups like aldehydes, ketones, Aromatic compounds etc., can be recognized.
- Biology point of view: Recognition of biomolecules, which can be micromolecules or macromolecules.
- Molecules with weight more than thousand dalton – **Biomacromolecules**.
- Molecules with weight less than thousand dalton – **Biomicromolecules**.

Biomicromolecules

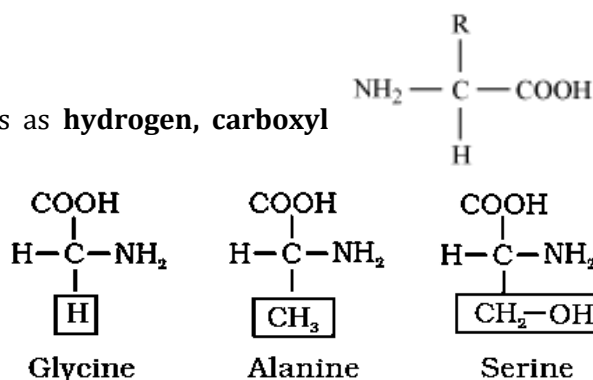
- Organic and inorganic constituents of living cells are classified as **amino acids, fatty acids, nucleotide bases**, etc.

Amino Acids

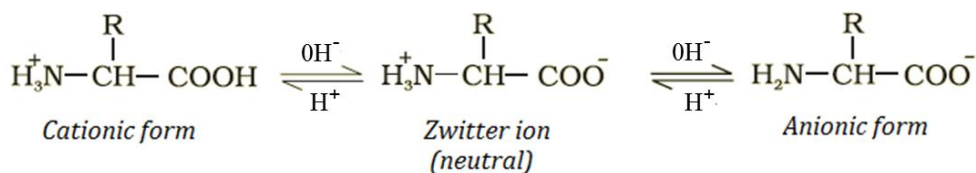
- In these compounds, α -carbon has the substituents as **hydrogen, carboxyl group, amino group** and a variable group **R**.

- Also called as **α -amino acids** because, functional groups are attached to same carbon.

- Based on the R group, there are **21 amino acids**
 - When R is H – **Glycine** (the simplest amino acid)
 - When R is CH_3 – **alanine**
 - When R is CH_2OH – **serine**



- The chemical and physical properties of amino acids depend upon their amino group, carboxyl group and R group.
- Based on the number of amino and carboxyl groups, amino acids could be
 - **Acidic amino acid** – contains two carboxylic group – Ex., Glutamic acid, Aspartic acid
 - **Basic amino acid** – contains two amino group – Ex., Lysine, Arginine.
 - **Neutral amino acid** – contains equal amino and carboxylic groups – Ex., Valine
- **Aromatic amino acids** – e.g., tyrosine, phenylalanine and tryptophan
- At different pH, the structures of amino acids change because of **ionisable nature** of $-\text{NH}_2$ and $-\text{COOH}$ groups.
- In neutral solution, an amino acid molecule exists as a dipolar ion (zwitter ion) i.e., molecule containing positive and negative ionic groups. The charge on the ion changes with pH. In an acidic solution (low pH) the amino acid picks up H^+ ions and becomes positively charged. In alkaline solution (high pH) the amino acid donates H^+ ions to the medium and becomes negatively charged. The pH at which the amino acid is electrically neutral or the molecule exists as zwitterion is called **isoelectric pH**.



- In neutral solution, an amino acid molecule exists as a dipolar ion (zwitter ion) i.e., molecule containing positive and negative ionic groups. The charge on the ion changes with pH. In an acidic solution (low pH) the amino acid picks up H^+ ions and becomes positively charged. In alkaline solution (high pH) the amino acid donates H^+ ions to the medium and becomes negatively charged. The pH at which the amino acid is electrically neutral or the molecule exists as zwitterion is called **isoelectric pH**.

Extra information – 20 amino acids and their three letter and one letter abbreviation

Essential Amino acids	Three letter abbreviation	One letter abbreviation	Non – essential amino acids	Three letter abbreviation	One letter abbreviation
Isoleucine	Ile	I	Alanine	Ala	A
Leucine	Leu	L	Asparagine	Asn	N
Lysine	Lys	K	Aspartic acid	Asp	D
Methionine	Met	M	Cysteine	Cys	C
Phenylalanine	Phe	F	Glutamic acid	Glu	E
Threonine	Thr	T	Glutamine	Gln	Q
Tryptophan	Trp	W	Glycine	Gly	G
Valine	Val	V	Proline	Pro	P
Arginine	Arg	R	Serine	Ser	S
Histidine	His	H	Tyrosine	Tyr	Y

Lipids

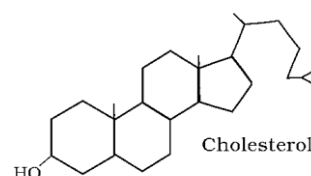
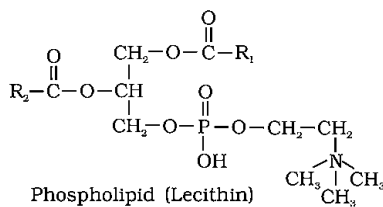
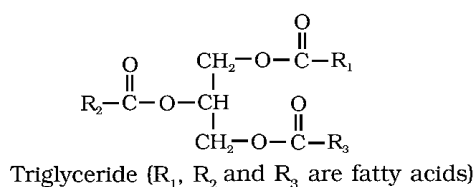
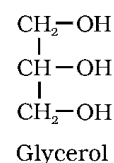
- Lipids are water insoluble
- Lipids could be fatty acids or glycerol or combination of both.

Fatty Acids

- Have carboxylic group linked to an R group; R (could be methyl ($-CH_3$), ethyl ($-C_2H_5$) or any $-CH_2$ group with 1 to 19 carbons)
- General formula:- **$COOH - R$**
- Example –
 - Palmitic acid** has **16 carbons** including carboxyl carbon found in palm oil, meat, cheeses, butter and dairy products. **$COOH - (CH_2)_{14} - CH_3$** .
 - Arachidonic acid** has **20 carbons** including carboxyl carbon. **$COOH - (CH_2)_{18} - CH_3$** . It is found in phospholipids of membranes of the body cells and is abundant in the brain.
- Fatty acids could be **saturated** (without double bond. Eg., Palmitic acid) or **unsaturated** (with one or more double bond ($C=C$). Eg., Arachidonic acid, Oleic acid, Linoleic acid, Linolenic acid)

Glycerol

- This is trihydroxy propane.
- It is a simple lipid.
- Many lipids have both fatty acids and glycerol.
- Here fatty acids are found esterified with glycerol
- Based on the number of fatty acids, lipids could be **monoglycerides, diglycerides, triglycerides**, etc.



- Some lipids are **phosphorylated** (conjugated lipid – on hydrolysis yield fatty acid, glycerol plus non lipid group like glucose, phosphorus); they contain phosphorus, e.g., **lecithin**
- Lipids are also called as Fats and Oils based on melting point
- Fats - have high melting point. E.g., Ghee, Butter, Vansapati
- Oil** – have low melting point and hence remain as oil in winter. E.g., Palm oil, Coconut oil, gingely oil

- **NOTE** - Other related compound on hydrolysis yield fatty acid, glycerol and alcohol other than glycerol which includes **Sphigolipids** (occurs in brain and nervous tissue), **Steroids** (occurs in hormones like testosterone, estrogen, progesterone) **Sterols** (occurs in cholesterol, Vit D, etc) and **Prostaglandins** (occurs in prostanoids)

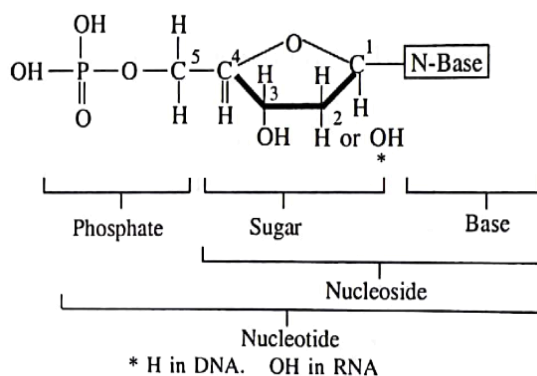
Nitrogenous Bases

- Nitrogen containing carbon compounds with heterocyclic rings.
- Nitrogen bases are classified into two types – **Purine** and **Pyrimidine**.
- **Purine** – consists of two ring structures and includes **Adenine (A)** and **Guanine (G)**.
- **Pyrimidine** – consists of single ring structure and includes **Thymine (T)**, **Uracil (U)** and **Cytosine (C)**.
- Nitrogen bases are also found attached to pentose sugar (ribose and deoxyribose) called **Nucleoside**.
- **Nucleosides** = Nitrogen bases + Sugar.
e.g., **adenosine, guanosine, thymidine, cytidine, uridine**.
- Nucleoside are also found attached to phosphate group called **Nucleotide**.
- **Nucleotides** = Nitrogen base + sugar + Phosphate group.

OR

Nucleotides = Nucleoside + Phosphate group

E.g., **Adenylic acid, Guanylic acid, Thymidylic acid, Cytidylic acid, Uridylic acid.**



- Nucleic acids like **DNA** and **RNA** are polymers of nucleotides.

Nitrogen bases	Nucleosides	Nucleotide
Adenine (Purine)	Adenosine	Adenylic acid
Uracil (Pyrimidine)	Uridine	Uridylic acid

Biomacromolecules

- The biomacromolecules which fall in the acid insoluble fraction include **polysaccharides, nucleic acids, proteins** and **lipids**.
- All biomacromolecules have molecular weight more than 10,000 Da (>1,000 Da), except lipids. This is because except lipids, all are polymeric substances.

NOTE - Lipids have a molecular weight of 800 Da, yet they come in the acid soluble fraction because they are water insoluble and are not polymeric substance. Hence lipids are not strictly macromolecules.

Table: Average Composition of Cells

Component	% of the total cellular mass
Water	70 – 90
Proteins	10 – 15
Carbohydrates	3
Lipids	2
Nucleic acids	5 – 7
Ions	1

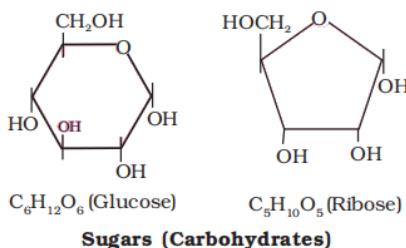
In the table given above the chemical composition of living tissue from abundance point of view and arrange them class wise, we observe that water is the most abundant chemical in living organisms

Polysaccharides

- Polysaccharides are long chains of sugars (monosaccharides).
- Polysaccharides can be **homopolysaccharides** or **heteropolysaccharides** of monosaccharides.

Homopolysaccharides

- Homopolysaccharides are the polysaccharides made up of one type of monomers.
- Example – Cellulose, starch, glycogen, inulin



Cellulose

- Cellulose is homopolymer of **β glucose** linked **β 1-4 glycosidic linkage**
- Structure of cellulose is unbranched and forms linear chain.
- Constitutes the plant cell wall, paper pulp, cotton fibre, etc.

Starch:

- Starch is homopolymer of **α glucose** linked by **α 1-4 glycosidic linkage**.
- Structure of starch is branched and forms helical secondary structures and can hold **I₂** molecules in the helical portion; the **starch-I₂ complex** is **blue** in colour, forming the basis of the **confirmatory test** to detect starch.
- Starch is storehouse of energy **in plants**.

Glycogen (animal starch)

- Glycogen is a homopolysaccharides of glucose.
- Found as a storage polysaccharide **in animals**.
- The right end is called the reducing end and the left end is called the non-reducing end.

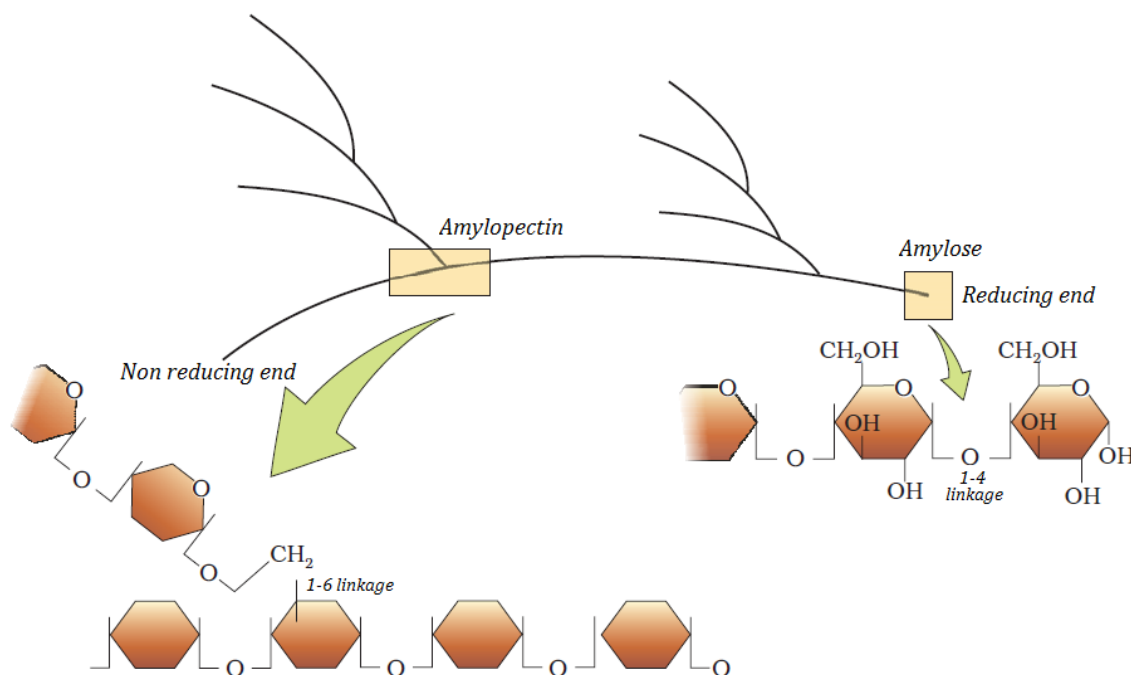


Figure: Diagrammatic representation of a portion of glycogen

Inulin: Polymer of fructose

Heteropolysaccharide

- Complex polysaccharide made up of amino-sugars and chemically modified sugars (e.g., glucosamine, N-acetyl galactosamine, etc.
- Example – Chitin (exoskeletons of arthropods and cell wall of fungi), Pectin

Nucleic Acid

- Nucleic acids are the polymers of nucleotides (Polynucleotide chain)
- Nucleotide = Nitrogenous bases (Heterocyclic compound) + Sugar + Phosphate group
- Nitrogenous bases** – Two types:
 - Purines** – Adenine (A) and Guanine (G)
 - Pyrimidines** – Thymine (T), Cytosine (C), Uracil (U)
- Sugar** – Two types:
 - Ribose** (forms RNA)
 - 2' deoxyribose** (forms DNA)
- A nucleic acid containing deoxyribose is called deoxyribonucleic acid (DNA).
- A nucleic acid containing ribose is called ribonucleic acid (RNA)

Proteins

- Proteins are polymers of amino acids.
- Proteins are polypeptides where amino acids are linked by peptide bonds in the form of a linear chain.

- Proteins are **heteropolymers** because the polypeptide chain consists of **different** types of amino acids.
- Amino acids – 2 types:
 - Essential** – which the body cannot make; supplied by diet (refer table given in page no.3)
 - Non-essential** – which the body can make (refer table given in page no.3)

Functions of proteins:

- Transportation of nutrients across membranes
- Fight infections (Antibodies)
- Act as hormones and enzymes

Protein	Function
Collagen	Intercellular ground substance
Trypsin	Enzyme
Insulin	Hormone
Antibody	Fights infectious agents
Receptor	Sensory reception (smell, taste, hearing, etc.)
GLUT – 4	Enables glucose transport into cells

- Collagen** – most abundant protein in the human body.
- RuBisCO** (Ribulose biphosphate Carboxylase-Oxygenase) – most abundant protein in the biosphere.

Structure of Proteins

- Structure Of Protein is Studied at Four Levels
 - Primary structure
 - Secondary structure
 - Tertiary structure
 - Quaternary structure

Primary Structure

- It refers simply to the positional information about the sequence of amino acid in a protein.
- The positional thread has amino acids arranged as a line, with the first amino acid as the N terminal amino acid and the last as the C terminal amino acid.

Secondary Structure

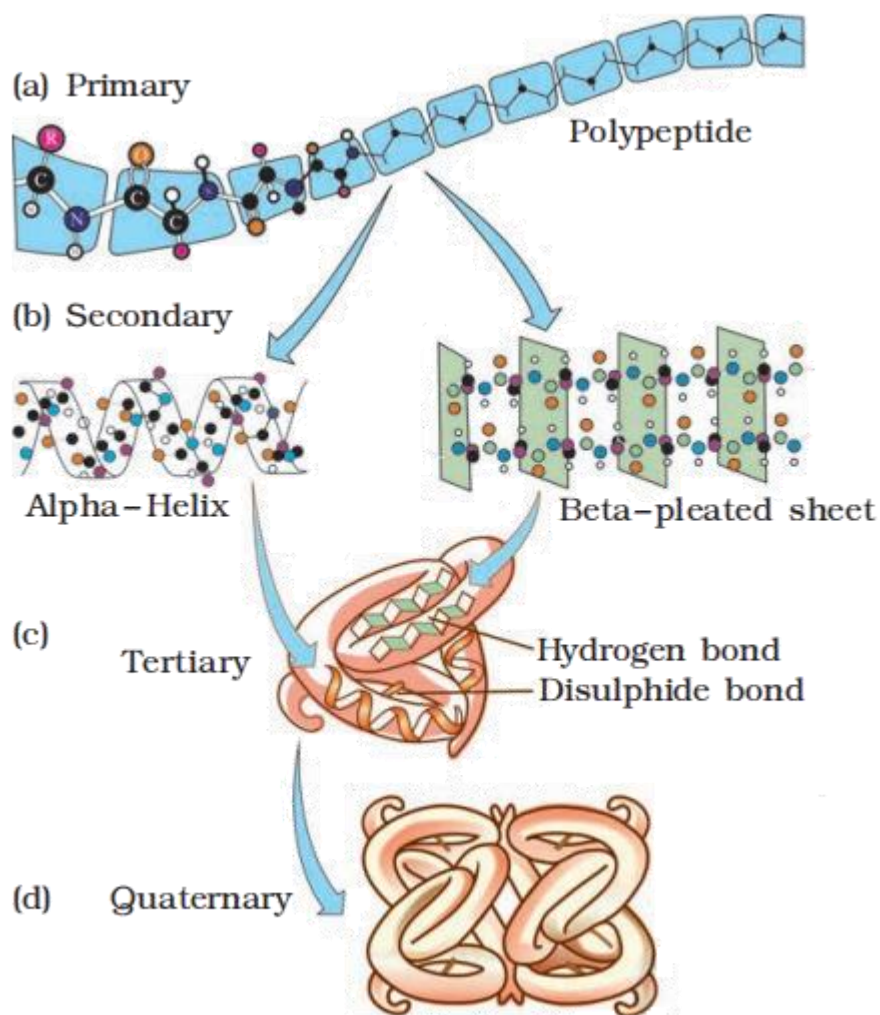
- Secondary structures are formed when the flexible positional thread folds to form structures like α helices and β sheets.

Tertiary Structure

- Tertiary structures are formed when the positional thread folds more intensively to form three-dimensional structures.
- Protein is biologically active at this stage.

Quaternary Structure

- Some proteins may have more than one assembly or sub-units.
- The manner in which these sub-units are arranged with respect to each other forms the quaternary structure of a protein.
- Example – Adult **haemoglobin** consists of 4 sub-units (2 α and 2 β)



Primary Metabolites – Intermediates or products of metabolism directly involved in growth, development, and reproduction. Example – fatty acids, amino acids, etc.

Secondary Metabolites – Intermediates or products of metabolism not involved directly in growth, development, and reproduction. Example – Pigments such as carotenoids, toxins, drugs, and essential oils

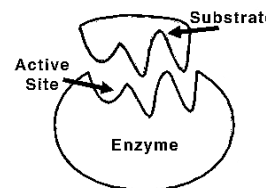
The roles of secondary metabolites are not directly identifiable. Some of them act as things for human utility such as **rubber, pigments, spices, drugs**, etc., while some others may have ecological importance.

Some Secondary metabolites

- Pigments – Carotenoids, Anthocyanines, etc
- Alkaloids – Morphine, Codeine, etc
- Terpenoids – Mono-, Di-terpenes, etc
- Essential oils – Lemon grass oil, etc
- Toxins – Abrin, Ricin
- Lectins – Concanavalin A
- Drugs – Vinblastin, curcumin, etc
- Polymeric substance – Rubber, gums, cellulose

Enzymes

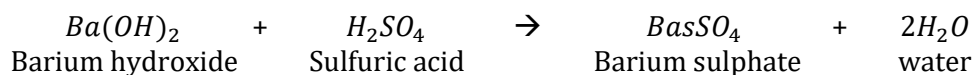
- Almost all enzymes are proteins, except ribozymes (nucleic acids acting like enzymes).
- Like all proteins, enzymes also have primary, secondary and tertiary structures. In tertiary structure, the protein (enzyme) folds in such a criss- cross way that many crevices or pockets are made. These crevices are called **active sites**.
- In an active site, the substrate fits in and reaction is catalysed at a high rate.
- Unlike inorganic catalysts, enzymes get damaged at high temperatures (above 40°C), except those isolated from thermophilic organisms (organisms living in hot vents and sulphur spring enzymes show activities upto the temperature $80^{\circ} - 90^{\circ}\text{C}$)



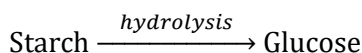
Chemical Reactions

- A change in substance could be **physical** (change in shape without breaking of bonds and change state of matter) or **chemical** (involves making and breaking of bonds).

Example 1 – Inorganic reaction



Example 2 – Organic reaction

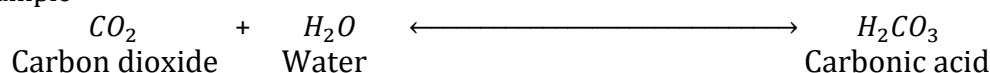


- Rate of reaction:** is defined as “Amount of product (P) formed per unit time (t).” And it is expressed as

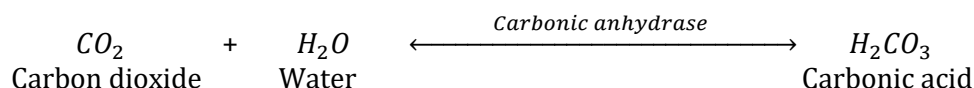
$$\text{Rate of reaction} = \frac{\delta P}{\delta t}$$

- Temperature directly influences the rate of reaction. Rate doubles or halves for every **10°C** change in either direction.

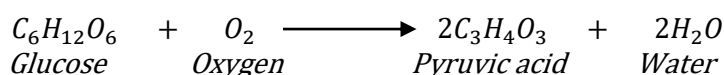
Example –



- In absence of enzymes this reaction is very slow with about 200 molecules of H_2CO_3 being formed in an hour. i.e., Rate of reaction = 200 molecules of H_2CO_3 / Hour.
- By using enzyme (carbonic anhydrase) present within the cytoplasm, the reaction speeds dramatically with about 6 00,000 molecules being formed every second. i.e., Rate of reaction = 6 00,000 molecules of H_2CO_3 / Sec.



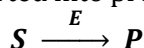
- Using a catalyst, the rate of reaction can be increased by about **10 million times** in certain cases.
- In a metabolic pathway, specific enzymes catalyse particular metabolic reactions.



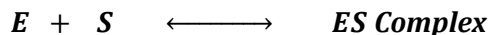
- It is actually a metabolic pathways in which one molecule of glucose becomes 2 molecules of pyruvic acid through 10 different enzyme catalysed metabolic reactions.

Process of Enzyme Action

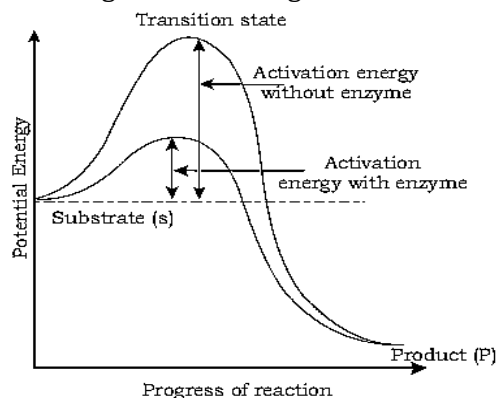
- Substrate (S) is the chemical that is converted into product (P) by the action of an enzyme (E).



- First of all, the substrate 'S' binds to the enzyme 'E' at its active site. This leads to the formation of the **Enzyme-Substrate 'ES' complex**.

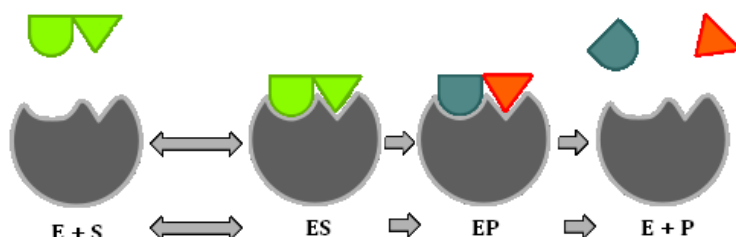


- Transition state** - structure is the new structure of the substrate being formed during the state when the substrate is bound to the enzyme. It is the transitional structure between the substrate and the product.
- Finally, the structure of the substrate gets converted into the structure of the product, and the product is released from the active site.
- Transition state is the state of higher energy and lesser stability as compared to the product.
- The amount of energy needed to get the transition state is called activation energy. Or the difference in average energy content of 'S' from its transition state is called **activation energy**.



Catalytic Cycle of Enzyme

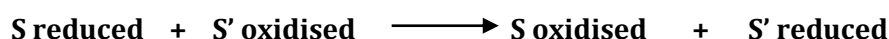
- Substrate binds to the active site of an enzyme, fitting into the active site.
- Shape of the enzyme is altered by the binding of the substrate, and it fits more tightly around the substrate.
- Bond between the enzyme and the substrate is broken and a new enzyme-product complex is formed.
- Product is released from the enzyme, and free enzyme is now ready to enter a new catalytic cycle by binding with a new substrate.



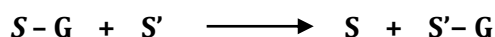
Classification and Nomenclature of Enzymes

- Enzymes are classified into **6 classes** based on the reactions they catalyse.
- These 6 classes, each having 4–13 subclasses, are named by four digits numbers.
- 6 classes of enzymes are as follows:

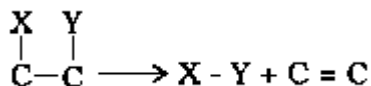
- Oxidoreductases / dehydrogenases** – catalyse oxidation and reduction (redox) reactions between two substrates



- Transferases** – catalyse the transfer of a group (G) between two substrates S and S'. G is any group other than hydrogen



3. **Hydrolases**– catalyse hydrolysis of bonds like ester, peptide, C – C, glycosidic, etc. with the help of H and OH group of water molecule (H₂O)
4. **Lyases** – catalyse the removal of groups from the substrate by mechanisms other than hydrolysis, leaving double bonds.



5. **Isomerases**– catalyse inter-conversion among various isomeric forms
6. **Ligases**– catalyse the linking together of 2 compounds; e.g., enzymes catalysing the joining of C – O, C – S, C – N bonds

NOTE – A catalytically active enzyme consists of two components namely apoenzyme and **cofactors** and is called as **Holoenzyme**

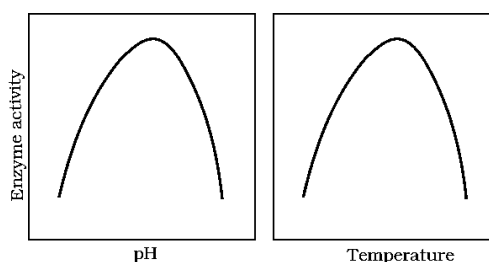
- **Apoenzyme** - Protein portion of an enzyme is called **apoenzyme**.
 - **Cofactors** are the **non-protein** components bound to enzymes.
 - Cofactors make enzymes catalytically active.
 - 3 types of co-factors: prosthetic groups, co-enzyme and metal ions
1. **Prosthetic group** – are the organic compounds and binds tightly to the apoenzyme; e.g., haem (part of the active site of an enzyme)
 2. **Co-enzyme** – are the organic compound and forms transient association with apoenzyme; occurs only during the course of catalysis; e.g., NAD and NADP
 3. **Metal ions** – these are inorganic and basically required to form coordination bonds with side chains of the active site and with the substrate; e.g., carboxypeptidase contains Zn²⁺

FACTORS AFFECTING ENZYME ACTIVITY

- Temperature
- pH
- Change in substrate concentration
- Binding of specific chemicals

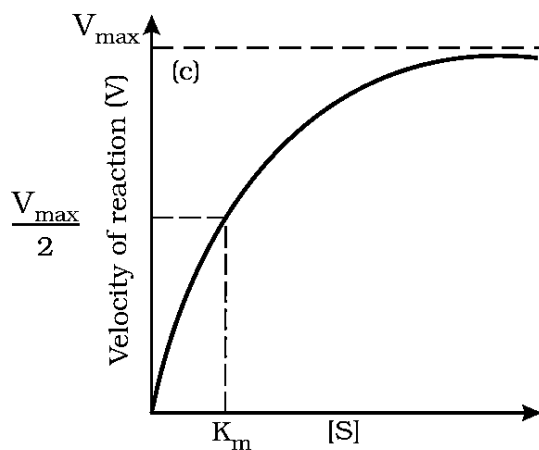
Temperature and pH

- Enzymes have a narrow range of temperature and pH optima. At this optimum temperature and optimum pressure, enzyme activity is the highest.
- Below and above this optima, the activity of an enzyme declines.
- High temperature – denatures an enzyme
- Low temperature – preserves an enzyme in inactive state



Concentration of Substrate

- With an increase in substrate concentration, the velocity of enzymatic reaction increases at a very fast rate, and then reaches its maximum velocity V_{max}
- Velocity does not increase beyond V_{max} even if concentration is increased further.
- This happens because enzyme molecules saturate with substrate molecules, and no enzyme molecules are left to bind with additional substrates.



Where,

- K_m is Michaelis menten constant
- Michaelis menten is defined as the substrate concentration at half the maximum velocity

Binding of Certain Chemicals

- Inhibitors are the chemicals that shut down enzymatic activity by a process called **inhibition**.
- **Competitive inhibition:** Here, the inhibitor closely resembles the substrate, and binds with the active sites of enzymes. Hence, enzyme activity declines.
- Example – malonate resembles succinate structurally, and blocks the activity of the enzyme succinic dehydrogenase